

FINAL EXPLANATION: BIOLOGY GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 10: SUB-STRAND 2.3 PLANT GASEOUS EXCHANGE AND RESPIRATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation. Use everything you learned in Lessons 1 to 12 to answer the driving question: How do plants "breathe," and what happens inside their cells when they release the energy stored in food? Anchor your answers in the sukuma wiki (kale) bubble phenomenon from Mwalimu Njoroge's class in Nairobi: freshly cut leaves in warm water (about 42°C) gave streams of bubbles from specific spots, while leaves in cold water (about 15°C) gave far fewer. Write in full sentences in the space under each prompt. Use scientific words correctly, give evidence from your own observations and investigations, and explain WHY, not only WHAT. You may draw and label diagrams where a prompt asks. Do not leave out the link between the cell level and the whole-leaf level.

Part 1: Where do the bubbles come from?

In Mwalimu Njoroge's class, bubbles streamed from specific spots on the submerged sukuma wiki leaf, not from the whole surface. (a) Name the structures on the leaf that these bubbles come from, and describe how they are built and where they are found on the leaf. (b) Describe what makes each of these structures change between letting gas through and not letting gas through, using one of the two theories you studied about how this happens. (c) Name the other gas-exchange sites in plants, and say where each is found.

(a) The bubbles come from stomata. Stomata are microscopic pores in the leaf epidermis, found mostly on the lower surface. Each pore is surrounded by two guard cells. Gas escapes in a stream at these spots because the stomata are the openings through which gases move in and out. (b) Potassium ion theory: K⁺ ions are actively pumped into the guard cells. This lowers the water potential inside them, so water enters by osmosis. The guard cells become turgid and change shape, which opens the pore. When K⁺ and water leave, the guard cells lose turgor and the pore closes. (The starch-sugar hypothesis also works: starch changes to soluble sugars in light, water enters by osmosis, and the guard cells swell.) Both theories rely on osmosis and turgor pressure. (c) Other sites: the waxy cuticle (limited exchange), lenticels on woody stems and branches, for example mango bark, and pneumatophores, the specialised roots of plants in waterlogged places such as the mangroves at Mida Creek.

Part 2: What is happening inside the cells?

(a) Write the word equation or symbol equation for aerobic

(a) Glucose + oxygen → carbon dioxide + water + energy. In symbols: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$. It happens in the mitochondria, after the first stage in the cytoplasm. (b) Glycolysis happens in the cytoplasm. The Krebs cycle happens in the

respiration and say where in the cell it happens. (b) Name the three stages of aerobic respiration and the part of the cell where each takes place. (c) Compare aerobic and anaerobic respiration in yeast: oxygen needed, products, and ATP yield per glucose. (d) Describe one Kenyan example where anaerobic respiration is useful.	mitochondrial matrix. Oxidative phosphorylation happens on the inner mitochondrial membrane. (c) Aerobic respiration needs oxygen. It produces carbon dioxide and water and gives about 36 to 38 ATP per glucose. Anaerobic respiration in yeast happens without oxygen in the cytoplasm. It produces ethanol and carbon dioxide ($C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 + \text{energy}$) and gives only 2 ATP per glucose. (d) Mandazi dough rises because yeast ferments sugar without oxygen and the carbon dioxide forms gas pockets. Other examples are souring of uji, brewing busaa, biogas digesters in Kisumu, and ethanol production. In waterlogged soil, plant roots may also respire anaerobically.
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Part 3: Why did warm water make more bubbles?

Compare the warm water (about 42°C) and cold water (about 15°C) results. (a) Explain, using enzymes, why temperature changed the number of bubbles. (b) Name the two gases that a submerged leaf could release, the process that makes each, and what each process needs. Then explain what you can and cannot be sure of about the gas in the bubbles you saw, given the tests you did. (c) Describe the evidence from your yeast investigations that supports your temperature explanation, and say which investigation tested temperature and which did not.	(a) Respiration and photosynthesis are controlled by enzymes. In warm water the enzyme molecules and their substrate molecules have more kinetic energy and collide more often, so more reactions succeed each second. More gas is produced and more bubbles appear. In cold water the enzymes work slowly, so little gas is made and few bubbles appear. (b) Carbon dioxide is made by cellular respiration in the mitochondria of mesophyll cells. Oxygen is made by photosynthesis in the chloroplasts, but this needs light, so it only happens if the submerged leaf is lit. Some CO₂ from respiration would also dissolve in the water, so fewer CO₂ bubbles would be seen than the amount actually made. We did not test the bubbles from the leaf (for example with limewater), so we cannot be certain of their contents. Our best explanation, based on the yeast evidence and what we know about leaf cells, is that the bubbles are mostly gas released from leaf cells and air spaces through the stomata, with CO₂ from respiration and, in light, O₂ from photosynthesis possibly contributing. (c) The Lesson 6 investigation tested temperature: yeast in warm sugar-cane juice produced significantly more CO₂ bubbles per minute than yeast in cold water, so respiration rate increased with warmth, matching the sukuma wiki pattern. The Lesson 4 limewater test compared aerobic and anaerobic conditions, not temperature, so it shows that respiration makes CO₂ but does not itself show the temperature effect (though warming also made the limewater turn milky faster). The balloon bottles with different yeast concentrations also inflated because of CO₂ from yeast, showing that more yeast gave faster gas production, but they did not vary temperature. So the Lesson 6 data support the idea that temperature controls respiration rate through enzyme activity.
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Part 4: Plants as a linked system

(a) Explain how photosynthesis, gaseous exchange and respiration are linked in a sukuma wiki plant. Say what each process supplies to the others. (b) Answer the misconception: 'Plants only	(a) Photosynthesis in the chloroplasts uses CO₂ and water to make glucose and oxygen. Respiration in the mitochondria uses that glucose and oxygen to release energy as ATP, and its products are CO₂ and water. The CO₂ and water made by respiration are raw materials that photosynthesis can use, and the glucose and oxygen made by photosynthesis are the substrates that respiration uses, so the gases cycle in both directions between the two processes. Gaseous exchange structures such as stomata and lenticels let the gases enter and leave the plant, and guard cells control the openings. (b) This is wrong. Plant cells respire all the time, day and night, to release energy from glucose. They photosynthesise only when light is available. Plants respire, and the CO₂ this produces leaves through the gas-exchange sites. (c) Xerophyte: acacia and cactus plants of the Rift Valley and Turkana
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photosynthesise; they do not respire.' (c) Choose one adaptation for gaseous exchange in a xerophyte or a hydrophyte from Kenya and explain how it helps the plant.	have sunken stomata and a thick cuticle, which reduce water loss while still allowing some gas exchange. Hydrophyte: water hyacinth from Lake Victoria has stomata on the upper, air-facing surface and aerenchyma, air-filled tissue that carries gases to submerged parts.
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Part 5: The final explanation

Write a scientific explanation of the anchoring phenomenon for a classmate who missed the unit. Your answer must explain (a) where the bubbles come from and why only from specific spots, (b) what the gases are and what makes them, (c) why warm water gave more bubbles than cold water, and (d) whether plants 'breathe' like animals and what this means at cell level. Finish by connecting the cell level to the whole-leaf level.	The bubbles come from stomata, microscopic pores in the leaf surface, which is why they appear at specific spots and not over the whole leaf. Guard cells open each pore when K⁺ ions enter and water follows by osmosis, making them turgid. The gas in the bubbles comes from the leaf's cells and air spaces and leaves through the open stomata. Two gases are made in the leaf: carbon dioxide, made by aerobic respiration in the mitochondria of mesophyll cells (glycolysis, Krebs cycle and oxidative phosphorylation, giving about 36 to 38 ATP per glucose), and oxygen, made by photosynthesis in the chloroplasts, but only if the submerged leaf receives light. Because some CO₂ dissolves in the water, the bubbles may contain relatively less CO₂ than the leaf produces. We did not test the leaf bubbles directly, so this is our best explanation from the evidence of the unit rather than a proven fact. Warm water (about 42°C) gave more bubbles than cold water (about 15°C) because higher temperature increases the kinetic energy of enzymes and substrates, so enzyme-controlled reactions such as respiration run faster and make more gas, as the warm and cold yeast results showed. Cold water slows the enzymes and reduces bubbles. Plants do 'breathe' in the sense that they continuously exchange gases with their environment through stomata, lenticels, the cuticle and pneumatophores, but they have no lungs. The real work is chemistry inside cells: respiration releases energy stored in glucose as ATP, the same chemistry as in all living cells, including those of a Rift Valley runner. If oxygen is missing, cells can release a smaller amount of energy (2 ATP per glucose) by anaerobic respiration. So a visible stream of bubbles from a sukuma wiki leaf is the large-scale sign of microscopic reactions in mitochondria and chloroplasts, with the stomata linking the cell level to the whole leaf.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Structures and mechanism of gas exchange	Names stomata, guard cells and the other sites correctly and explains opening and closing using osmosis, turgor and K ⁺ or starch-sugar, linked to the bubble spots.	Names stomata and most sites and gives a basic account of guard cell opening, with small gaps in the mechanism.	Names few sites or says bubbles come from the whole leaf; no mechanism for how pores open.
Cellular respiration content	Gives correct equations, location of each stage, ATP yields (36 to 38 versus 2) and compares aerobic and anaerobic products accurately.	Gives correct equations and some locations or ATP figures, with minor errors.	Confuses the products or conditions of aerobic and anaerobic respiration, or omits equations.
Use of evidence and enzyme reasoning for temperature	Uses data from the sukuma wiki and yeast investigations, matches each investigation to what it actually tested, is honest about what was not tested, and explains the temperature effect through	States that warmth speeds up reactions and cites some evidence, with limited reasoning.	States the temperature effect without evidence or reasoning.

	enzyme kinetic energy and collisions.		
Linking photosynthesis, respiration and gas exchange	Clearly shows the two-way flow of CO ₂ , water, O ₂ and glucose, corrects the misconception, and links the cell level to the whole leaf.	Describes the link between the processes, with the cell-to-leaf connection only partly explained.	Treats processes as separate or keeps the misconception that plants do not respire.
Scientific communication and Kenyan context	Writes a clear, well-organised, accurate narrative using correct terms and Kenyan examples.	Writes clearly with mostly correct terms and some local examples.	Writing is unclear or disorganised, with incorrect terms and no examples.

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